

CHAPTER ONE

PREDICTION OF STOCK MARKET USING MACHINE LEARNING TECHNIQUES

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

The stock market is a vital component of the financial system that enables companies to raise capital while providing investors with opportunities to earn returns through equity ownership. Stock prices are influenced by several factors, including company performance, macroeconomic conditions, political stability, industry trends, and investor expectations. Over time, two major approaches have been traditionally used to analyze and predict stock market behavior: **fundamental analysis**, which evaluates a company's intrinsic value based on financial statements, economic indicators, and news events; and **technical analysis**, which focuses on historical price movements and trading volumes to identify patterns and trends that may indicate future price behavior (*Patel et al., 2015; Adebisi et al., 2014*).

With the advancement of computational techniques, machine learning has emerged as a powerful tool for stock market prediction due to its ability to model complex, nonlinear relationships within large datasets. Unlike traditional statistical methods, machine learning algorithms can automatically learn patterns from historical data and adapt to changing market conditions. However, studies have shown that models relying solely on technical indicators or historical prices often fail to capture external influences such as economic news, corporate announcements, and investor sentiment, which play a crucial role in market movements (*IJARCCCE, 2022; ResearchGate, 2018*).

Recent research has therefore shifted toward integrating qualitative information with quantitative market data to enhance predictive accuracy. Sentiment analysis of news articles, financial reports, and macroeconomic narratives has been shown to improve forecasting performance by capturing the psychological and informational drivers of investor behavior. This integration allows predictive systems to respond not only to past price trends but also to real-world events that directly affect market confidence and valuation (*IJRASET, 2023; Frontiers in Artificial Intelligence, 2025*).

The emergence of **Large Language Models (LLMs)** has further strengthened this hybrid approach by enabling advanced natural language processing of unstructured financial text. LLMs can extract deeper semantic meaning, contextual sentiment, and latent patterns from news headlines, economic reports, and political developments. Studies indicate that combining LLM-based textual feature extraction with traditional technical indicators leads to more robust and adaptable stock market prediction systems compared to conventional sentiment analysis techniques (*Frontiers in Artificial Intelligence, 2025*).

Despite the growing body of research on hybrid stock market prediction models, most existing studies focus on developed markets such as the United States, Europe, and Asia. Emerging markets like Nigeria remain under-represented in the literature, even though they exhibit higher volatility, sensitivity to political and economic changes, and rapid information diffusion. Factors such as exchange rate instability, oil price dependency, regulatory changes, and political events significantly influence the Nigerian Stock Exchange, making it a suitable but challenging environment for advanced predictive modeling (*IJARCCCE, 2022*).

In response to these gaps, this study proposes a hybrid stock market prediction framework that integrates **fundamental analysis**, **technical analysis**, and **Large Language Models** to forecast future stock prices of companies listed on the Nigerian Stock Exchange. By leveraging both numerical market data and rich textual information, the proposed approach aims to improve prediction accuracy and provide a more comprehensive understanding of stock price movements within the Nigerian financial market context (*Frontiers in Artificial Intelligence*, 2025; *ResearchGate*, 2018).

1.2 Statement of the Problem

Predicting stock market prices remains a complex and challenging task due to the volatile, dynamic, and information-sensitive nature of financial markets. Stock prices are influenced by numerous interrelated factors such as historical price movements, company financial performance, macroeconomic conditions, political events, and investor sentiment. Traditional forecasting approaches often fail to capture the full scope of these influences, resulting in inaccurate or inconsistent predictions, especially during periods of high market uncertainty (*Patel et al.*, 2015; *Adebiyi et al.*, 2014).

Many existing stock market prediction models rely heavily on **technical analysis**, using historical price and volume data to infer future price movements. While such models are effective in identifying short-term trends and patterns, they largely ignore external information such as news events, policy changes, and economic indicators that significantly impact investor behavior. As a result, technically driven models often perform poorly when unexpected market events occur, limiting their reliability in real-world trading environments (*IJARCCCE*, 2022; *ResearchGate*, 2018).

Conversely, models that incorporate **fundamental analysis** frequently struggle with the challenge of processing unstructured textual data, such as news articles, corporate disclosures, and political reports. Traditional sentiment analysis techniques often lack the contextual understanding required to interpret complex financial narratives accurately. This limitation reduces the effectiveness of fundamental-based models, as they may misinterpret sentiment or fail to capture subtle semantic relationships within financial text (*IJRASET, 2023*).

Although recent studies have attempted to combine fundamental and technical analysis using machine learning techniques, many of these approaches utilize shallow natural language processing methods that do not fully exploit the richness of textual information. The emergence of **Large Language Models (LLMs)** provides a more advanced means of extracting semantic meaning and sentiment from financial text; however, their application in stock market prediction remains relatively underexplored, particularly in emerging markets (*Frontiers in Artificial Intelligence, 2025*).

Furthermore, existing Large Language Models often suffer from 'contextual blindness' when applied to the Nigerian financial ecosystem. Most models are pre-trained on Western financial corpora, making them less sensitive to local market-moving narratives such as Naira redesign policies, fuel subsidy removal nuances, or the specific volatility patterns of the NGX. This study addresses the lack of a localized hybrid framework that can interpret the unique socio-economic sentiment driving the Nigerian stock market. (*IJARCCCE, 2022*).

Therefore, the problem addressed in this study is the absence of a robust and comprehensive stock market prediction system that effectively integrates **technical indicators**, **fundamental information**, and **advanced Large Language Models** to predict stock price movements within

the Nigerian stock market. Addressing this problem is essential to improving prediction accuracy, supporting informed investment decisions, and advancing research on hybrid financial forecasting models tailored to emerging market environments (*Frontiers in Artificial Intelligence*, 2025; *ResearchGate*, 2018).

1.3 Aim and Objectives of the Study

The **main aim** of this study is to develop a hybrid machine learning forecasting model that integrates fundamental (textual and economic) and technical indicators using Large Language Models (LLMs) to predict future stock prices of companies listed on the Nigerian Stock Exchange.

To achieve this aim, the study will pursue the following **specific objectives** to:

1. Collect and preprocess historical stock price data and fundamental signals (news, financial reports) for selected Nigerian companies.
2. Extract meaningful features from textual data using advanced LLM NLP techniques.
3. Compute relevant technical indicators from historical price data.
4. Train a hybrid machine learning model that integrates extracted fundamental features and technical indicators.
5. Evaluate the predictive performance of the proposed model against baseline technical-only and fundamental-only models.
6. Analyze how the inclusion of fundamental signals affects prediction accuracy and robustness in Nigerian stock forecasting.

1.4 Significance of the Study

This study has **both academic and practical significance**:

- **Advancing Research:** It contributes to the limited literature on hybrid forecasting models that combine technical data and deep semantic analysis of textual information for emerging markets like Nigeria. Unlike most models focusing solely on developed markets, this study addresses a gap by situating research within a *dynamic, information-sensitive, emerging stock market context*.
- **Improving Prediction Quality:** By integrating sentiment and macroeconomic factors with price history using state-of-the-art LLMs, the model aims to improve forecast accuracy compared to models restricted to technical indicators alone.
- **Investor Decision Support:** Investors and financial analysts can use the model to make more informed trading strategies that account for both market momentum and narrative shifts, especially important during volatile events like currency shocks, political changes, or earnings surprises.
- **Policy and Economic Insights:** For regulators and policymakers, understanding how fundamentals and market sentiment drive stock performance may inform policy decisions that stabilize markets or support investor confidence.

1.5 Scope of the Study

The study focuses on predictive modeling for selected publicly traded Nigerian companies listed on the **NGX 30 Index**. These companies are selected based on high liquidity and market capitalization, ensuring sufficient historical price data and a consistent stream of news for LLM processing.

The project does **not** attempt to forecast exact future price values over long horizons (e.g., multi-year forecasts) but rather to improve near-term trend prediction performance (short-to-medium term). The methodology also assumes access to timely text sources (e.g., news APIs) and reliable market price feeds.

1.6 Organization of the Study

This project is structured into five chapters:

Chapter Two – Comprehensive literature review contextualizing hybrid fundamental and technical forecasting models and current research trends.

Chapter Three – Research methodology outlining dataset sources, feature extraction, hybrid modeling approach, and evaluation framework.

Chapter Four – Implementation and results presentation, including model training, validation, and comparative performance analysis.

Chapter Five – Summary, conclusion, and recommendations for future research and application in stock market forecasting.

CHAPTER TWO

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LITERATURE REVIEW

2.1 Introduction

Stock market prediction remains one of the most researched problems in finance and computer science because stock prices influence investment decisions, corporate financing, and national economic development. The ability to forecast future stock movements can support better portfolio management, reduce exposure to losses, and improve decision-making for investors and market regulators. However, predicting stock prices is difficult because financial markets are highly volatile and respond quickly to new information, making price patterns unstable over time. (*Patel et al., 2015*)

Historically, stock market forecasting began with traditional statistical and econometric methods such as regression analysis and time-series models. These approaches were useful in analyzing historical trends and relationships but were often limited by strong assumptions such as linearity, stationarity, and stable relationships between variables. In real markets, price movements are frequently nonlinear, influenced by sudden shocks, and driven by complex dependencies that standard statistical tools may fail to capture effectively. (*Adebiyi et al., 2014*)

The increasing availability of large financial datasets and improved computing power has accelerated the use of machine learning (ML) in stock market prediction. Machine learning models can learn complex patterns from data without being explicitly programmed and have been widely adopted for predicting stock price direction and trend behavior. Many studies report that ML models such as Support Vector Machines, Random Forests, and Neural Networks often

outperform traditional methods, especially when trained on engineered technical indicators derived from historical price and volume data. (*Kadam et al., 2022*)

Despite the success of technical-based machine learning approaches, there is increasing evidence that relying only on historical prices is insufficient because stock prices are not influenced solely by past trends. Real-world stock movement is driven by broader macroeconomic conditions, company financial health, regulatory developments, and political events. As a result, researchers increasingly recommend hybrid approaches that combine technical indicators with fundamental data to capture both market dynamics and underlying economic drivers that influence valuation and investor behavior. (*Ogiemudia, 2022*)

In addition, the rapid growth of digital information sources has made financial markets more information-driven than ever. News releases, company announcements, policy decisions, and public sentiment can trigger rapid changes in investor behavior, causing immediate market reactions. To capture these qualitative market drivers, modern forecasting studies increasingly integrate Natural Language Processing (NLP) techniques that extract sentiment signals from financial news and reports to enhance predictive systems. (*IJARCCE, 2022*)

Recent advances in Large Language Models (LLMs) have further strengthened sentiment-driven forecasting because they can interpret text contextually and extract deeper semantic meaning beyond simple positive or negative scoring. Hybrid frameworks combining technical indicators with LLM-derived textual features are gaining attention because they improve responsiveness to event-driven market shocks, which is especially important in emerging markets like Nigeria where political and macroeconomic instability can rapidly influence stock movement. This study therefore aligns with modern research by focusing on a Nigeria-centered hybrid LLM forecasting

model that integrates both technical and fundamental information for improved stock market prediction. (Rohan et al., 2025; Araci, 2019)

This study is grounded in the **Adaptive Market Hypothesis (AMH)**, which suggests that markets are not always efficient but evolve based on investor behavior and environmental changes. In the context of the Nigerian Stock Exchange, which often exhibits 'weak-form' efficiency, machine learning models can exploit non-linear patterns and information lags that traditional models overlook.

2.2 Overview of the Stock Market

The stock market is an organized financial system where shares of publicly listed companies are bought and sold. It supports capital formation by enabling firms to raise funds from investors while offering opportunities for investors to earn returns through dividends and capital appreciation. The stock market also supports liquidity, allowing investors to convert shares into cash through trading in a regulated environment. (Kadam et al., 2022)

Stock markets are typically divided into the primary market (where new shares are issued) and the secondary market (where existing shares are traded among investors). Price discovery occurs mainly in the secondary market as investors react to information and adjust their demand and supply decisions. In most markets, price movement is influenced by the interaction of rational evaluation and behavioral reactions to information. (Adsure et al., 2023)

Nigeria's stock market is represented by the Nigerian Exchange Group (NGX), which provides trading infrastructure and supports listing and market operations. As an emerging market, Nigeria's market behavior is often characterized by higher sensitivity to macroeconomic

conditions, currency pressures, policy shifts, and liquidity constraints compared to many developed markets. These realities make Nigeria a meaningful and challenging case for applying hybrid prediction models. (*Ogiemudia, 2022*)

Stock price movements are driven by firm-level fundamentals (profitability, debt levels, governance, sector conditions), macroeconomic indicators (inflation, interest rates, exchange rate), and event-driven information such as regulatory changes and political developments. In modern markets, the speed and scale of information diffusion through news outlets and digital media also play a major role in shaping short-term price dynamics. (*Rohan et al., 2025*)

2.3 Fundamental Analysis in Stock Market Prediction

Fundamental analysis focuses on evaluating a company's intrinsic value using its financial health and the economic environment. It considers variables such as earnings, revenue growth, balance sheet strength, debt exposure, dividend policy, sector performance, and macroeconomic stability. The central assumption is that over time, market prices tend to move toward intrinsic value, meaning mispricing creates investment opportunities. (*Kadam et al., 2022*)

Macroeconomic indicators such as inflation, interest rate policy, and exchange rates are frequently cited as major drivers of stock market performance, especially in emerging economies. In Nigeria, exchange rate and oil-linked macroeconomic dynamics are often highlighted as important in determining stock market returns and volatility patterns, reinforcing the importance of including macro variables in prediction models. (*Ogiemudia, 2022*)

Fundamental analysis today increasingly includes textual and narrative information such as financial news, corporate disclosures, policy announcements, and political events. Because these

sources are unstructured, modern forecasting research often uses NLP and LLM techniques to transform text into features that can be combined with numerical market data for prediction.

(Rohan et al., 2025)

2.4 Technical Analysis in Stock Market Prediction

Technical analysis involves the study of historical price and trading volume to identify patterns that may help forecast future price movement. Common techniques include trend-following indicators such as moving averages and momentum indicators such as RSI and MACD. These indicators are widely used because they provide structured numerical signals that reflect investor behavior embedded in price action. *(Kadam et al., 2022)*

Many machine learning studies treat technical analysis indicators as engineered features and feed them into models such as regression learners, SVM, Random Forest, and deep networks. Comparative experiments suggest that deep learning models like LSTM can outperform some classical approaches in certain datasets due to their ability to learn time dependencies, although results may vary significantly with data quality and evaluation design. *(Bansal et al., 2022)*

However, a consistent limitation in technical-only prediction is vulnerability to sudden events and regime shifts. Since technical indicators depend on historical patterns, abrupt changes caused by major news, policy shocks, or sentiment shifts can reduce predictive reliability. This motivates hybrid models that integrate technical indicators with fundamentals and text-derived signals. *(Saberironaghi et al., 2025)*

2.5 Risk Factors Affecting Stock Market Trends

Risk factors refer to market conditions and uncertainties that influence volatility, returns, and predictability. These factors are particularly important in stock prediction because they can introduce instability into time-series patterns and reduce the generalization performance of predictive models. (*Kadam et al., 2022*)

The major risk factors include the following:

- **Market Risk (Systematic Risk):** This is the risk that affects the entire stock market due to macroeconomic shocks such as inflation spikes, recessions, global crises, and widespread investor fear. Systematic risk cannot be eliminated through diversification and typically causes broad price declines across sectors. (*Saberironaghi et al., 2025*)
- **Exchange Rate Risk:** Exchange rate instability affects foreign investor participation, import costs, and firm profitability especially for firms with foreign-denominated obligations or reliance on imported inputs. Studies focusing on Nigeria identify exchange rate risk as a key determinant of stock returns and market volatility. (*Ogiemudia, 2022*)

Interest Rate and Monetary Policy Risk: Interest rate changes influence borrowing costs, corporate expansion, investor portfolio decisions, and valuation discount rates. Monetary policy tightening can reduce stock market attractiveness relative to fixed-income returns, triggering reallocation and price adjustments. (*Ogiemudia, 2022*)

- **Oil Price and Commodity Risk:** Since Nigeria's macroeconomic performance is highly linked to oil revenue, oil price shocks can influence exchange rate pressures, fiscal stability, and overall investor confidence. These macro linkages make oil price risk relevant to Nigerian stock market performance and volatility. (*Ogiemudia, 2022*)
- **Political and Regulatory Risk:** Political instability, elections, changes in regulation, and uncertainty around policy direction can increase risk premiums and alter investor sentiment. For emerging markets, this can cause abrupt shifts in price behavior that technical models may struggle to capture without fundamental context. (*Ogiemudia, 2022*)
- **Liquidity Risk:** Liquidity risk arises when markets have limited trading depth, causing large transactions to move prices sharply. Lower liquidity increases volatility and makes trend prediction more difficult, particularly for short-term forecasting. (*Rohan et al., 2025*)
- **Company-Specific Risk (Unsystematic Risk):** This relates to firm-level issues such as poor earnings, weak governance, legal disputes, management failure, or operational disruption. Company-specific risk can dominate price movement even when macro factors appear stable, making firm fundamentals essential in modeling. (*Kadam et al., 2022*)
- **Information and Sentiment Risk:** News shocks, rumors, and shifts in investor mood can trigger rapid buying or selling. Because modern markets react quickly to information, sentiment extracted from news and digital content can significantly influence short-term movement, supporting the inclusion of NLP/LLM features in prediction systems. (*Rohan et al., 2025*)

2.6 Machine Learning Approaches to Stock Market Prediction

Deep learning techniques like **Long Short-Term Memory (LSTM)** networks and other recurrent architectures are widely used for time-series forecasting due to their ability to capture temporal dependencies in sequential data (*Gil et al., 2024*). Other advanced architectures (e.g., Transformers, Temporal Fusion) are gaining attention for trend prediction tasks, offering improved modeling of complex patterns compared to traditional methods (*Gil et al., 2024*).

However, deep models require careful **hyperparameter tuning**, larger datasets, and significant computational resources. Without proper design and regularization, they risk overfitting to training data and generalize poorly during real-world prediction (*MDPI, 2025*).

2.7 LLMs and Hybrid Models in Stock Market Prediction

Hybrid models integrate multiple data types, commonly technical indicators from price history and fundamental drivers such as macroeconomic indicators and text-derived sentiment. The key advantage of hybrid approaches is that they capture both internal market structure (technical signals) and external event-driven forces (fundamental and sentiment signals). This improves robustness, especially during turbulent periods. (*Saberironaghi et al., 2025*)

Large Language Models strengthen hybrid modeling because they can process large-scale unstructured financial text and represent meaning using contextual embeddings. Unlike

traditional sentiment approaches that rely on fixed dictionaries, LLM-based modeling captures complex language structures, uncertainty tone, and nuanced sentiment. This is important in finance because market-moving signals may appear in subtle narratives rather than simple positive/negative words. *(Rohan et al., 2025)*

Domain-specific language models such as FinBERT were introduced to improve financial sentiment analysis by adapting transformer models to financial corpora and tasks. This supports the argument that general-purpose NLP models may misinterpret financial language, and domain tuning improves performance in extracting sentiment signals relevant for market prediction. *(Araci, 2019)*

In the Nigerian context, a hybrid LLM-based approach is particularly relevant because the market is sensitive to macroeconomic instability, policy shifts, exchange rate pressures, and political events that strongly shape investor confidence. Integrating technical indicators with text-based fundamental features creates a more comprehensive framework for predicting Nigerian stock trends compared to technical-only pipelines. *(Ogiemudia, 2022)*

2.7 Model Evaluation Metrics

Model performance is typically assessed via **classification accuracy**, precision, recall, F1-score, and error metrics like RMSE, MAE, and MSE. Comparative studies indicate that no single model universally outperforms others; performance depends on dataset characteristics, feature engineering quality, and choice of evaluation metric *(Wagh & Gaikwad, 2025)*.

Ensemble and neural models often attract attention for higher predictive capability, but simpler models can outperform them in specific scenarios, especially with careful feature selection and data preprocessing (Li, n.d.)

2.8 Related Works / Empirical Studies

This section reviews empirical studies and case works closely related to stock price/trend prediction using **statistical methods, machine learning, deep learning (LSTM), and hybrid approaches that combine market data with textual/sentiment information**. The purpose is to show what researchers have already done, the methods they used, the kind of data they worked with, and the key outcomes that inform the direction of this project.

Egbunu, Yange, and Idoko (2023) investigated the prediction of stock market returns in Nigeria using a Long Short-Term Memory (LSTM) recurrent neural network. Their work used historical stock prices spanning **2 January 2014 to 22 September 2021**, applied data extraction on the price series, and compared the proposed LSTM approach against ANN and a Fuzzy-GA model using error-based evaluation. Their results reported that the LSTM approach achieved lower prediction errors (MSE/RMSE) than the competing baselines, supporting the suitability of LSTM for modeling nonlinear and volatile return series in the Nigerian context. (*Egbunu et al., 2023*)

Hingane, Bhate, Patil, and Ukhalkar (2025) provided a structured literature review focused on stock price prediction with machine learning approaches and statistical methods, emphasizing how price forecasting is affected by uncertainty and “shock” behavior in markets. Their review

synthesizes the common methodological pipeline seen across studies of data collection, preprocessing, feature extraction, model training, and evaluation and highlights that model performance often depends strongly on data quality, the choice of features, and the validation design. This study is relevant because it supports the need for multi-source modeling (technical + information signals) and careful experimental structure in stock prediction projects. (*Hingane et al., 2025*)

Chen (2022) presented a comparative discussion of several machine learning models for stock market prediction, including Support Vector Machine (SVM), regression-based models, LSTM, and CNN-based approaches. The study emphasizes that market prediction models should be assessed comparatively because performance can vary by model type and data properties, and it notes that hybrid deep learning combinations (e.g., CNN + LSTM) can perform strongly in some prediction settings. This work is relevant to the present study because it supports benchmarking multiple models and highlights that external drivers (e.g., politics, economic environment) can influence investor sentiment and market movement. (*Chen, 2022*)

Saberironaghi, Ren, and Saberironaghi (2025) published a broad review of machine learning and deep learning techniques for stock market prediction, covering commonly used algorithms such as SVM, CNN, and LSTM, along with recurring challenges like interpretability, data quality issues, and the difficulty of real-time response to fast-changing market conditions. Their review is relevant because it reinforces why hybrid systems increasingly integrate multiple data sources (historical prices + external signals) and why evaluation should go beyond a single metric to provide credible performance interpretation. (*Saberironaghi et al., 2025*)

Vuong, Phu, Nguyen, Duy, Bao, and Trinh (2024) conducted a bibliometric and methodological review tracing stock price forecasting from classical statistical models to modern machine learning/deep learning and hybrid frameworks. They emphasize that statistical methods such as ARIMA remain useful but face limitations in non-linear settings, while machine learning improves non-linear pattern learning, and deep learning captures complex temporal structures in price sequences. Their review also points out that hybrid models combine approaches to counterbalance the weaknesses of individual methods, which aligns with this project's motivation to combine technical and fundamental signals. (*Vuong et al., 2024*)

Gupta, Madan, Singh, and Singh (2022) proposed a forecasting model that integrates both historical data and sentiment information, arguing that news publication correlates strongly with price movement and that combining sentiment with historical patterns improves predictive accuracy. Their approach illustrates the broader hybrid direction where textual signals complement price-derived features, supporting the design choice in this project to include fundamental/news information rather than relying only on technical indicators. (*Gupta et al., 2022*)

Across these related works, a consistent pattern emerges: (i) purely statistical models often struggle with market nonlinearity and sudden regime shifts; (ii) machine learning improves predictive capability when strong engineered features are available; (iii) deep learning especially LSTM shows strong performance for sequential/temporal modeling; and (iv) hybrid approaches that integrate textual/sentiment information with price/technical data often improve robustness during event-driven volatility. A critical observation from the reviewed literature is the **temporal alignment of data**. While technical data is instantaneous, the impact of news sentiment often has a 'lag effect'. This study improves on existing works by implementing a time-weighted sentiment

window to capture how news released during non-trading hours (weekends and holidays) influences the next opening bell on the NGX. These findings directly justify the direction of this project combining technical time-series learning (LSTM-driven) with fundamental text understanding (LLM-driven) in the Nigerian stock market context. (*Egbunu et al., 2023; Hingane et al., 2025; Saberironaghi et al., 2025*)

CHAPTER THREE

CHAPTER THREE

RESEARCH METHODOLOGY

3.0 Introduction

This chapter presents the research methodology adopted for the project titled “**Stock Market Trend Prediction Using Machine Learning Models (Nigeria Case Study)**”. The proposed solution is a **hybrid prediction framework** that combines:

1. **Technical Analysis Modeling (Quantitative Time-Series):** Using historical Nigeria stock price data (OHLCV) to compute trend/momentum/volatility signals and train a deep learning model (LSTM) for prediction.
2. **Fundamental Analysis Modeling (Qualitative + Financial Signals):** Using a **Large Language Model (LLM)** to interpret and score fundamental drivers such as market news, company financial health indicators, economic events, and political/regulatory developments.
3. **Fusion Layer (Hybrid Decision):** Combining outputs from both technical and fundamental pipelines to predict the next-day/next-window stock direction and/or future price movement.

This approach reflects real market practice where investors consider both price behavior (technical) and real-world drivers (fundamental) before making decisions.

3.1 Research Design

The research adopts an **experimental design** using machine learning and deep learning approaches. The project is implemented as a **predictive system** consisting of data collection, preprocessing, feature engineering, model training, evaluation, and deployment.

The methodology is organized into the following phases:

1. **Data Acquisition** (NGX historical price data + news/fundamental sources)
2. **Data Preprocessing** (cleaning, normalization, feature alignment)
3. **Technical Feature Engineering** (returns and indicators)
4. **Fundamental Feature Engineering using LLM** (sentiment, event impact scoring, financial health summarization)
5. **Model Training**
 - LSTM-based technical model
 - LLM-based fundamental model
 - Hybrid fusion model
6. **Model Evaluation** (MSE, RMSE, classification metrics where needed)
7. **System Implementation** (architecture, modules, interfaces)

3.2 Study Area / Case Study: Nigerian Stock Market

This project focuses on stocks traded on the **Nigerian Exchange (NGX)**. The NGX provides publicly available equity price lists and daily trading information, which makes it suitable for building and evaluating predictive systems.

Example stocks that can be used include high-liquidity equities such as banks, cement/manufacturing, and consumer goods firms (e.g., DANGCEM, GTCO, ZENITHBANK, ACCESSCORP). The choice of Nigeria is important because local market dynamics differ from developed markets due to liquidity constraints, macroeconomic sensitivity, policy effects, and information diffusion rate.

3.3 Data Requirements and Sources

3.3.1 Technical Analysis Dataset (Quantitative)

The technical dataset is built from **historical price movement data**, typically containing:

- **Open price (O)**
- **High price (H)**
- **Low price (L)**
- **Close price (C)**
- **Volume (V)** (optional)

Primary sources include:

- NGX equities price list and historical market updates
- Financial market feeds where Nigeria stocks are tracked such as assessworth, stock bubbles (as backup)

3.3.2 Fundamental Analysis Dataset (Qualitative/Financial)

The fundamental dataset includes:

- **Market news headlines and full articles** (e.g., company news, earnings releases, macroeconomic news, FX policy news, inflation reports)
- **Company fundamentals** (financial health cues like profit trend, debt levels, cash flow commentary)
- **Macro and political signals** (elections, policy shifts, interest rate changes, regulatory actions)

These sources are processed by an **LLM** to extract structured numerical signals usable in a machine learning pipeline.

3.4 System Architecture of the Proposed Hybrid Model

3.4.1 High-Level Architecture

The proposed system is structured into major modules:

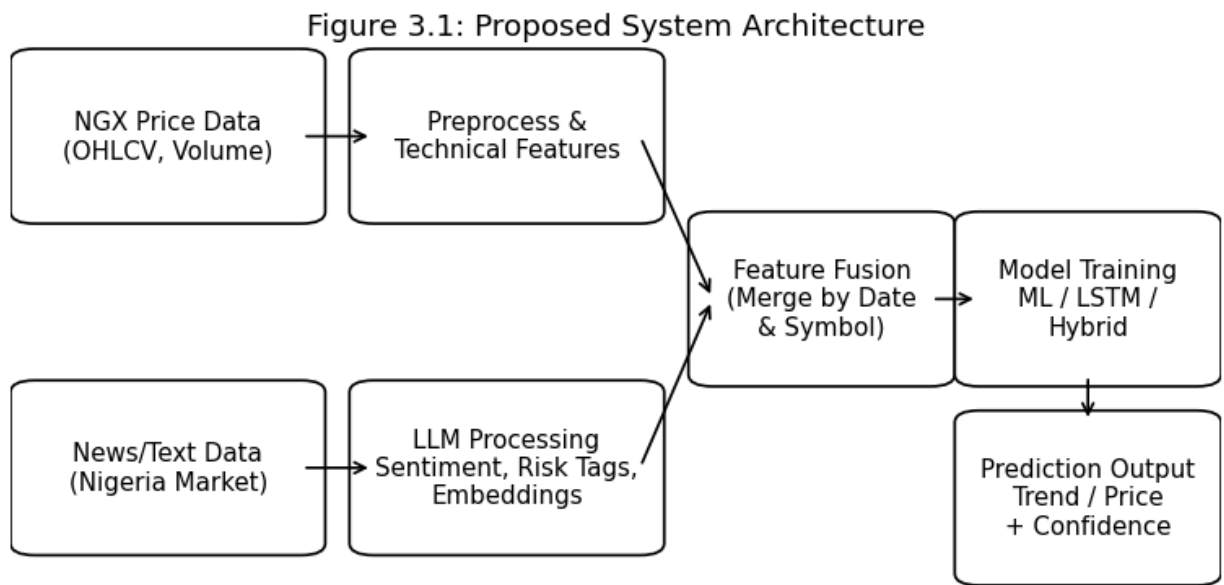
1. **Data Collection Module**
2. **Preprocessing and Feature Engineering Module**
3. **Technical Prediction Module (LSTM-based)**
4. **Fundamental Prediction Module (LLM-based)**
5. **Fusion & Decision Module (Hybrid Model)**

6. Evaluation Module

7. User Interface / Output Module

3.4.2 The Weekend & Holiday Alignment Logic

To handle the mismatch between 24/7 news cycles and 5-day trading weeks, a **Sentiment Aggregation Logic** is applied. Any news headline or report released between Friday's market close and Monday's market open is aggregated and assigned to Monday's trading day. This ensures the model accounts for 'overnight' information shocks that cause price gaps at the market open..



3.5 Technical Analysis Methodology (LSTM-Based)

This section adopts the LSTM time-series forecasting methodology used for predicting Nigeria stock returns, where historical stock prices are used to model future movement.

3.5.1 Technical Data Collection and Splitting

Historical OHLC data is collected for the selected NGX stock(s). The dataset is split into:

- **Training set:** 70%
- **Testing set:** 30%

This split ensures the model learns from the past while being tested on unseen data.

3.5.2 Technical Feature Engineering and Calculations

(a) Daily Return Calculation

A common way to represent price movement is through **returns**.

Simple return can be computed as:

$$r_t = \frac{C_t - C_{t-1}}{C_{t-1}}$$

Where:

- C_t = closing price at day t
- C_{t-1} = closing price at day $t - 1$

(b) Technical Indicators

To deepen predictive power, additional indicators are computed from OHLCV:

Trend indicators

- Simple Moving Average (SMA)

$$SMA_n = \frac{1}{n} \sum_{i=0}^{n-1} C_{t-i}$$

- Exponential Moving Average (EMA)

Momentum indicators

- Relative Strength Index (RSI)
- Rate of Change (ROC)

Volatility indicators

- Rolling standard deviation (σ)
- Bollinger Bands:

$$Upper = SMA_n + k\sigma, \quad Lower = SMA_n - k\sigma$$

- Average True Range (ATR)

3.5.3 LSTM Model Design (Technical Prediction Model)

(a) Concept of LSTM

LSTM is a recurrent neural network designed to model time dependencies using memory cells and gates.

The LSTM unit contains:

- **Forget gate**
- **Input gate**
- **Candidate cell state**

- **Output gate**

The **cell state update** is expressed as:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

And the hidden output:

$$h_t = o_t \odot \tanh(C_t)$$

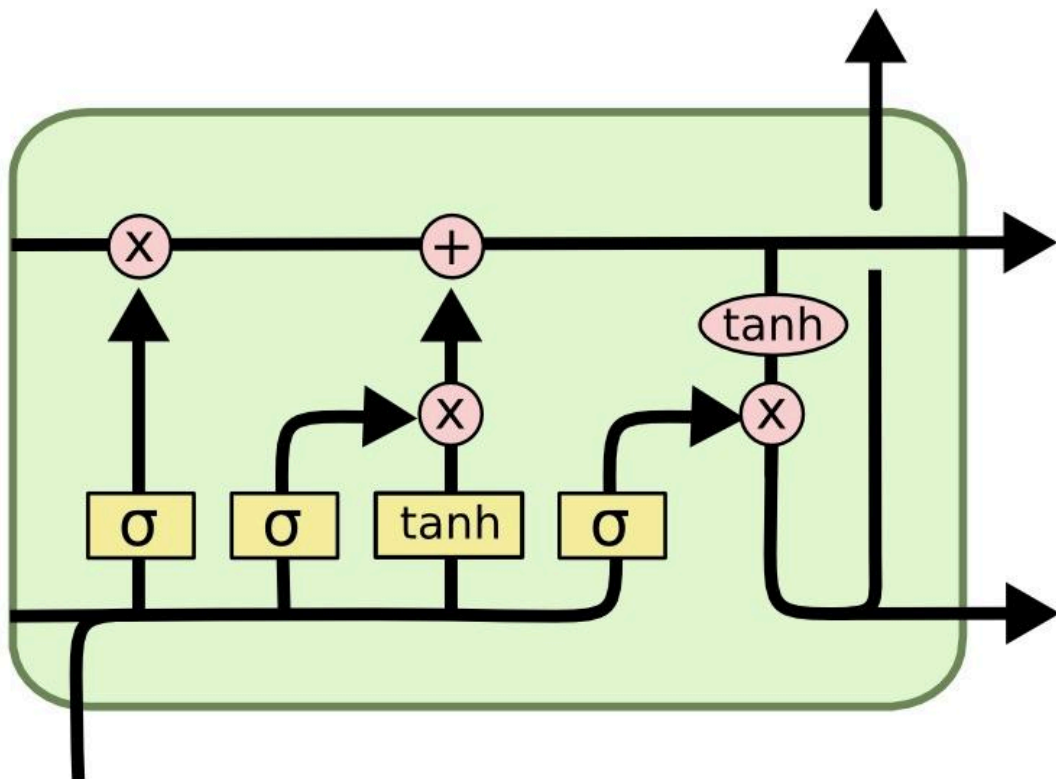


Figure 3.5:
Schematic Diagram of LSTM

3.5.4 Technical Model Target (Prediction Objective)

Two prediction formats are supported:

1. **Regression-based:** Predict the next closing price C_{t+1}
2. **Classification-based:** Predict the direction of return:
 - Positive return (Up)
 - Negative return (Down)

3.6 Fundamental Analysis Methodology (LLM-Based)

Technical indicators alone may miss major market-moving events (policy changes, earnings shocks, political news, FX volatility). To capture these, the system uses an LLM to interpret fundamental signals.

3.6.1 LLM Input Sources

The LLM receives:

- News headlines + article bodies
- Macro/political information (e.g., interest rates, inflation, FX policy)
- Company financial summaries (earnings reports, debt announcements, dividend news)

3.6.2 Fundamental Signal Extraction (LLM Tasks)

The LLM is used to convert unstructured information into structured features such as:

1. **Sentiment score** (negative/neutral/positive; or -1 to +1)
2. **Event impact score** (low/medium/high impact)
3. **Sector relevance tag** (banking, industrial, consumer goods, etc.)
4. **Company-specific risk flags** (e.g., “profit drop”, “high debt”, “regulatory risk”)
5. **Macro factor tags** (inflation shock, FX pressure, policy tightening)

3.6.3 Fundamental Feature Output

The output of the LLM becomes a numeric vector such as:

$$F_t = [sentiment_t, impact_t, policy_risk_t, earnings_signal_t, FX_signal_t]$$

This fundamental vector is aligned by date with the technical features.

3.6.4 LLM Feature Engineering: Deterministic Prompting

To ensure the reliability of fundamental features, the LLM is configured with a **Temperature setting of 0.0** to ensure deterministic (consistent) outputs. The extraction follows a strict JSON schema:

- **Sentiment:** Normalized scale of -\$1\$ (Bearish) to \$+1\$ (Bullish).
- **Significance:** \$0\$ to \$1\$ (impact on stock price).
- **Category:** Binary flags for Macro, Earnings, or Regulatory news.

3.7 Hybrid Fusion Model (Technical + Fundamental)

After producing:

- Technical feature vector T_t from indicators and LSTM hidden representations
- Fundamental feature vector F_t from LLM extraction

A fusion approach is applied:

3.7.1 Feature-Level Fusion

Combine both into one final vector:

$$X_t = [T_t \parallel F_t]$$

Then train a final classifier/regressor such as:

- Logistic Regression / Random Forest (for direction)
- XGBoost
- Feedforward neural network

3.7.2 Decision-Level Fusion (Alternative)

Predict separately:

- LSTM gives technical prediction P^{tech}
- LLM gives fundamental prediction P^{fund}

Then combine using weighted average:

$$P^{final} = \alpha P^{tech} + (1 - \alpha) P^{fund}$$

3.8 System Algorithm / Workflow

The system follows a step-by-step workflow consistent with the LSTM-based methodology:

1. Collect historical stock price data (CSV).
2. Select input variables (OHLCV).
3. Compute technical indicators and returns.
4. Collect fundamental news and financial texts by date.
5. Use LLM to extract structured fundamental signals.
6. Split dataset into training and testing sets.
7. Train the LSTM model on technical features.
8. Train fusion model (or combine predictions).
9. Evaluate performance using chosen metrics.
10. Output predictions and visualizations.

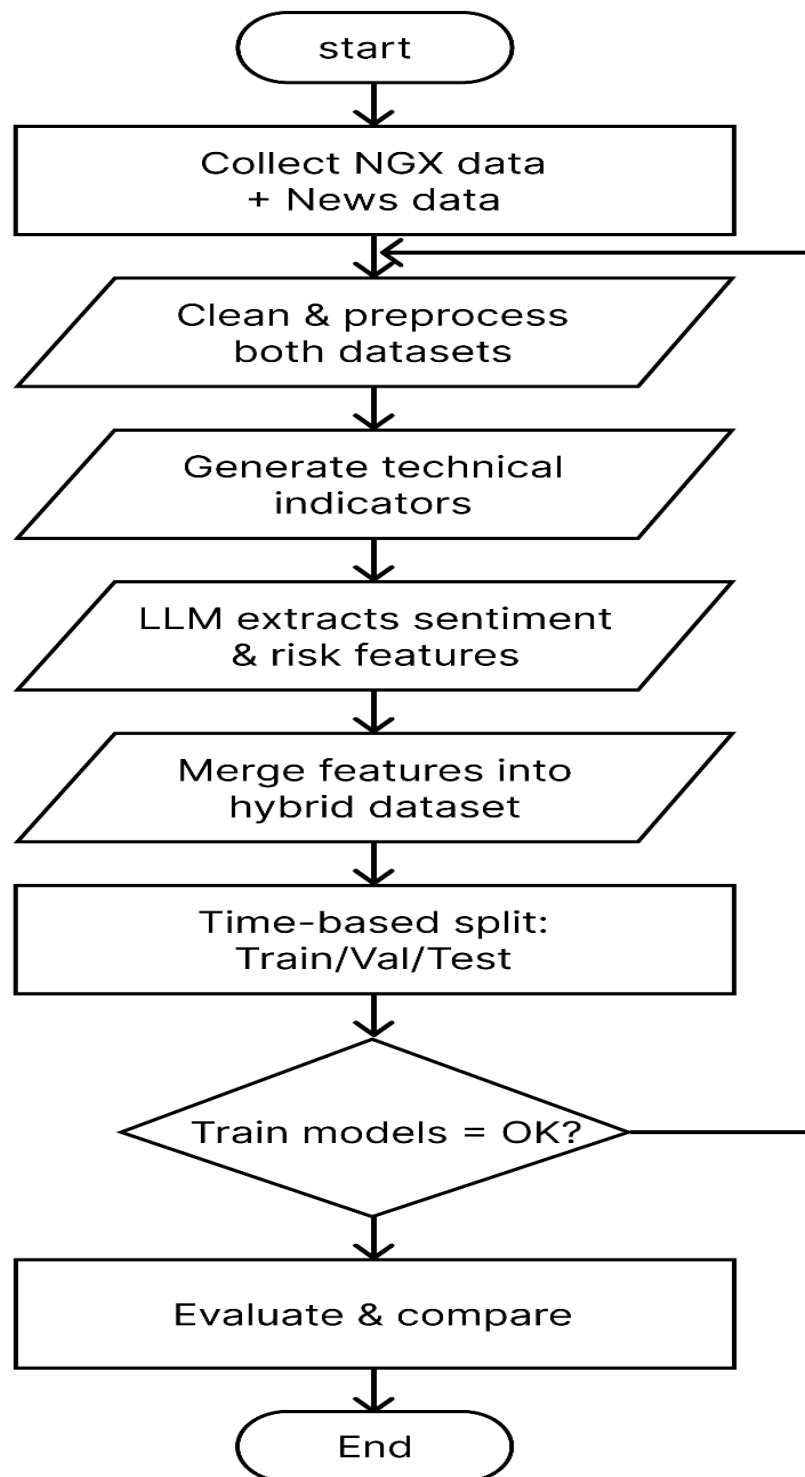


Figure 3.8: Methodology Flowchart

3.9 Model Evaluation Metrics

The model's performance is evaluated based on its dual output:

1. **For Price Prediction (Regression):** Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) are used to measure the deviation from actual closing prices.
2. **For Trend Prediction (Classification):** Accuracy, Precision, Recall, and the **F1-Score** are utilized to determine the model's ability to correctly identify 'Buy' (Up) and 'Sell' (Down) signals.

3.9.1 Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n \left(t_i - \hat{t}_i \right)^2$$

3.9.2 Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(t_i - \hat{t}_i \right)^2}$$

Where:

- t_i = actual value
- $\hat{t}_i$ = predicted value
- n = number of samples

3.10 System Design Diagrams (UML)

3.10.1 Use Case Diagram

Actors:

- User/Investor
- Admin/Developer

Key use cases:

- Upload/select stock
- Fetch market data
- Fetch news/fundamental data
- Run prediction
- View charts + prediction results
- Export report (optional)

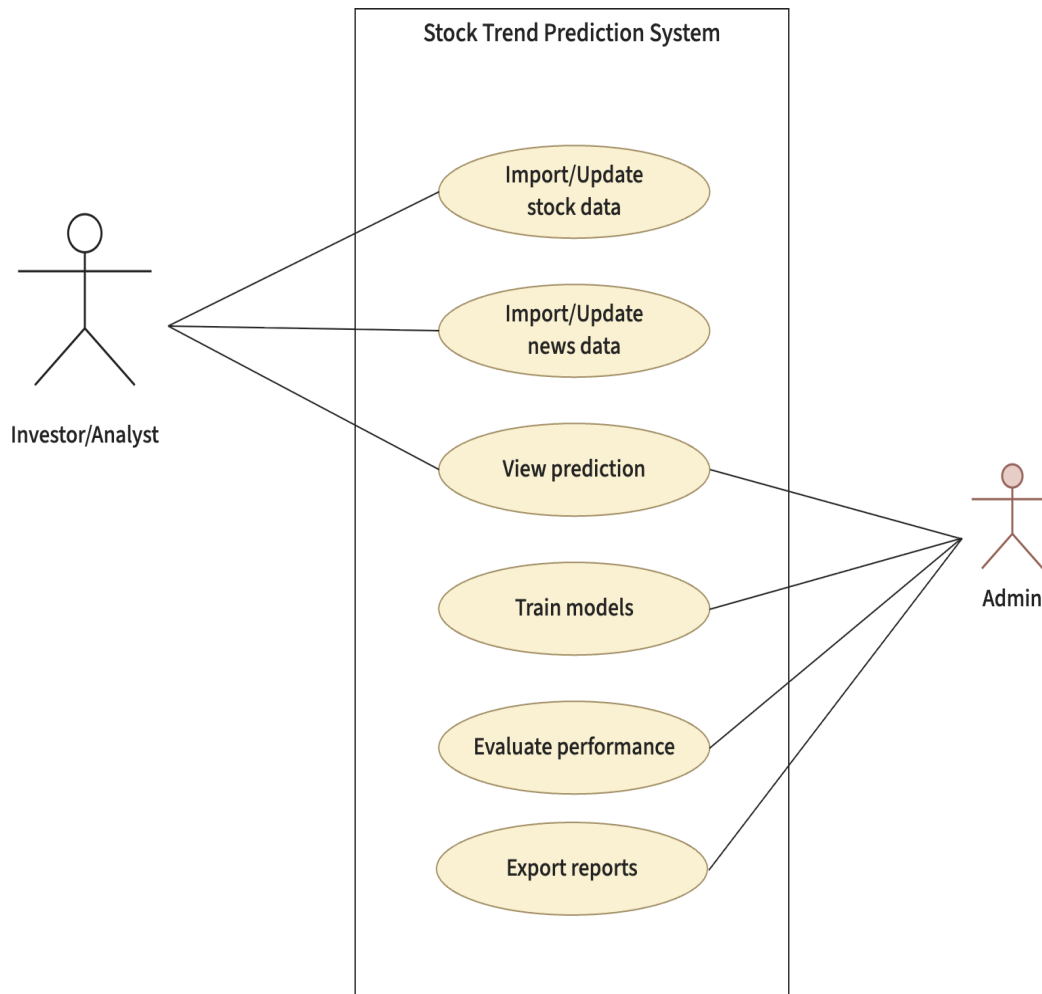


Figure 3.10.1: Use Case Diagram

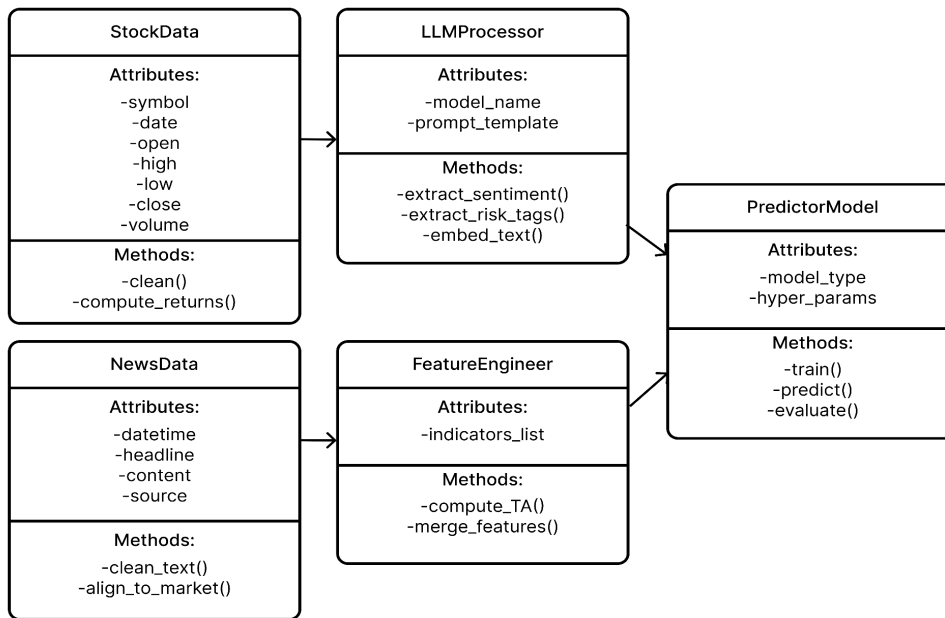
3.10.2 Class Diagram

Core classes include:

- DataCollector
- Preprocessor

- TechnicalIndicatorEngine
- LSTMModelTrainer
- LLMFundamentalAnalyzer
- FusionModel
- Evaluator
- PredictionService

Figure 3.10.2: Class Diagram



3.10.3 Sequence Diagram

Shows flow from user request → data retrieval → preprocessing → model inference → output.

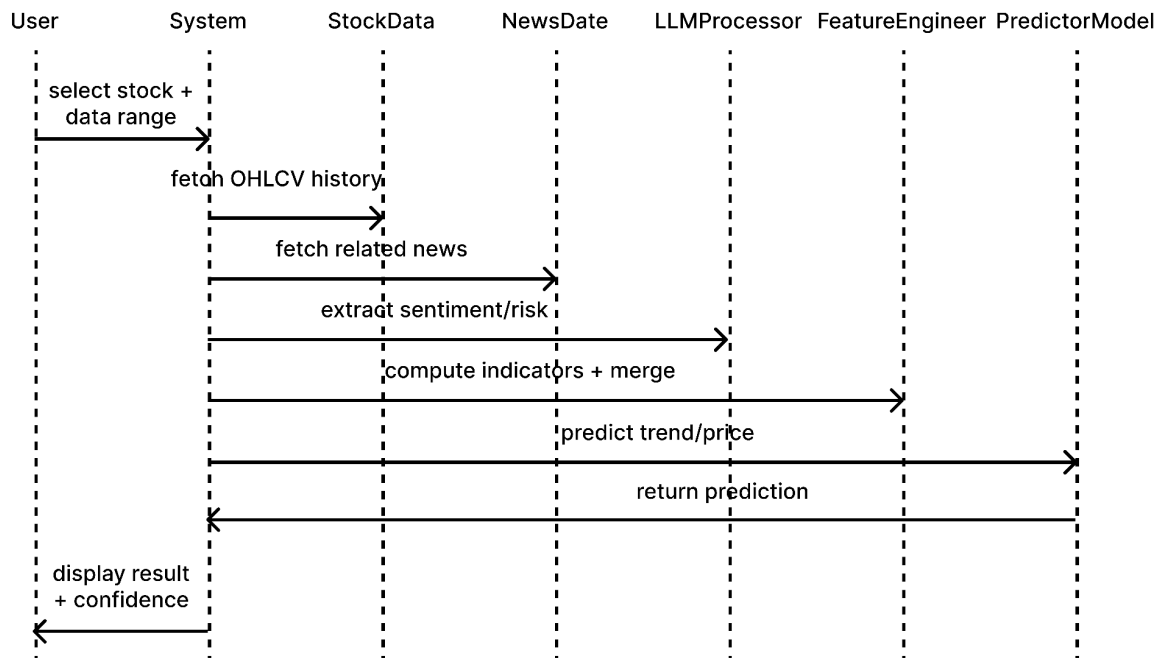


Figure 3.10.3:

Sequence Diagram (Prediction Request Flow)

3.11 Implementation Tools / Environment

The system will be implemented using:

- **Programming language:** Python
- **Libraries:** pandas, numpy, scikit-learn, TensorFlow/Keras or PyTorch
- **NLP/LLM tools:** OpenAI API / Llama / other transformer-based LLM pipeline
- **Visualization:** matplotlib, plotly
- **Development environment:** Jupyter Notebook / Google Colab

3.12 Summary

This chapter presented a complete methodology for a **hybrid stock prediction system** using both **technical** and **fundamental** analysis. The technical aspect uses an LSTM-based predictive approach built on historical price data and technical indicator computations, while the fundamental aspect uses an LLM to transform news and financial information into structured predictive signals. Both sources are fused to produce a final Nigeria-stock prediction output, which will be implemented and evaluated in the next chapter.

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